Workshop 6: Using Databases in Real Life

2026-04-22

Contents

In this workshop, you will work in a team on a table with many other students around using your databases in real life applications (web pages). It is a collective discussion to find a solution to the given problem, step by step by exchange and comments. Please use the MariaDB database with Docker as explained in lecture 00 (at the end). Please be positive, let all the people speak around the table, do not speak too loud, accept the comments for others, even negative comments can be treated as a way to improve your solution (always be positive). You can ask questions to me at any moment during the session. Välkomna, welcome, bienvenue...

1 Be positive! (5 min.)

Because, perhaps, you don't know each other around the table, take one minute per person to present yourself, explain who you are, where you come from, why you are taking this course/program and what you want to do in your future professional life. If everyone around the table is speaking Swedish, you can do it in Swedish, else, please go back to English.

2 Create the Person database (25 min.)

Please runMariadb in Docker as explained in lecture 00 (at the end) and create a database named "persondb" using the given code:

```
CREATE DATABASE IF NOT EXISTS persondb;
USE persondb;
CREATE TABLE IF NOT EXISTS Person (
  pid INT AUTO_INCREMENT PRIMARY KEY,
  pfirstname VARCHAR(100), plastname VARCHAR(100),
  pemail VARCHAR(255), pbirthdate DATE
);
INSERT INTO Person (pid, pfirstname, plastname, pbirthdate, pemail) VALUES
(1, 'John', 'Doe', '1990-01-01', 'john.doe@example.com'),(2, 'Jane', 'Smith', '1985-05-15',
'jane.smith@example.com'),(3, 'Alice', 'Johnson', '1992-09-30', 'alice.johnson@example.com'),(4,
'Bob', 'Brown', '1988-12-20', 'bob.brown@example.com'),(5, 'Charlie', 'Davis', '1995-03-10',
'charlie.davis@example.com'),(6, 'Emily', 'Wilson', '1991-07-25', 'emily.wilson@example.com'),(7,
'David', 'Miller', '1987-11-05', 'david.miller@example.com'),(8, 'Sarah', 'Taylor', '1993-02-18',
'sarah.taylor@example.com'),
// (9, 'Michael', 'Anderson', '1989-08-12', 'michael.anderson@example.com'),(10, 'Olivia', 'Thomas',
'1994-04-22', 'olivia.thomas@example.com'),(11, 'James', 'Jackson', '1986-10-30',
'james.jackson@example.com'),(12, 'Sophia', 'White', '1990-06-15', 'sophia.white@example.com');
```

3 Code the CRUD using Node.js (60 min.)

- The architecture we want to build to deliver a web page with the CRUD operations (Create, Read, Update, Delete) for the Person table to the final users is described in Figure 1.

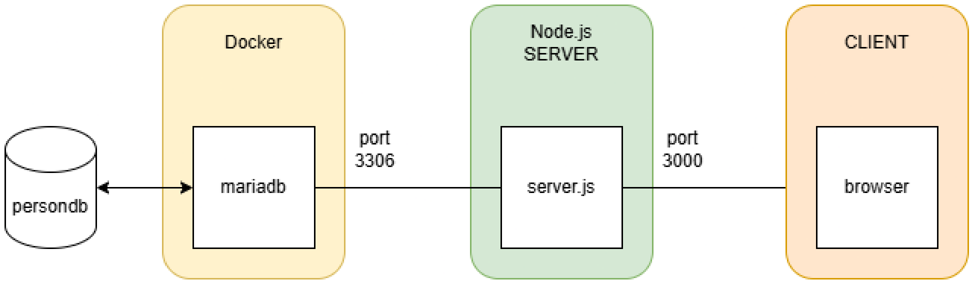
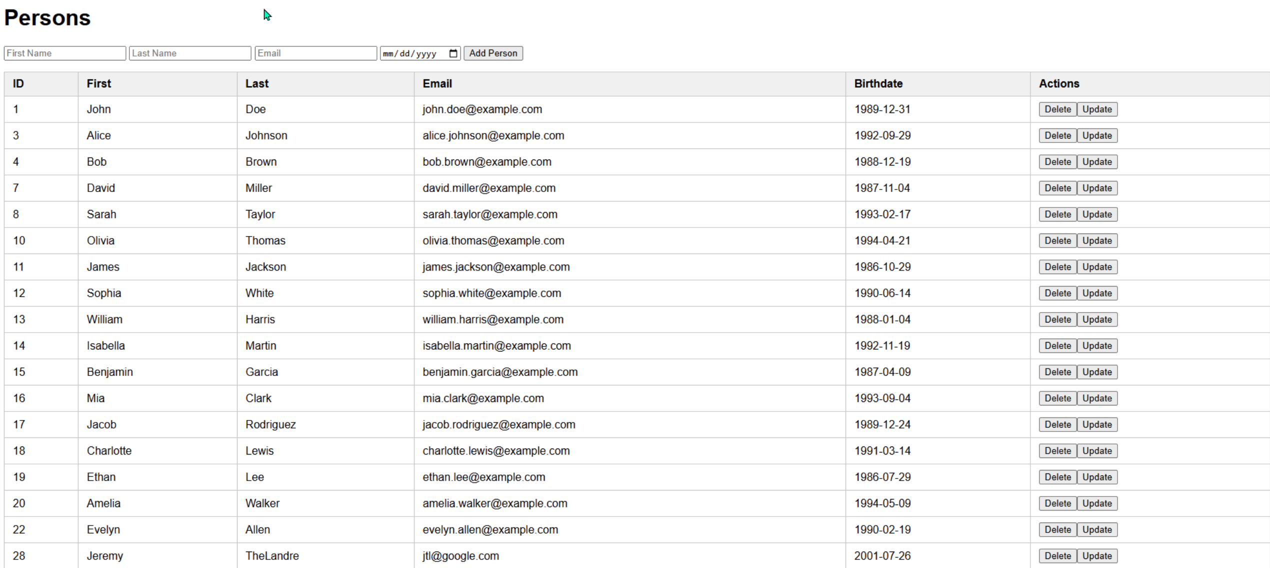


Figure 1: The web architecture.

- Procedure to start a new Node.js project from a terminal:
  1. Create a directory for the Node.js server: `mkdir workshop6`
  2. Go to the directory: `cd workshop6`
  3. Initialize a new Node.js project (this command will create a package.json file containing a description of your Node.js project): `npm init -y`
  4. Install the necessary dependencies (express and mysql2): `npm install express mysql2`
  5. Create a new file named “server.js” in the “workshop6” directory
  6. Create a new file named “data.html” in the “workshop6” directory
  7. Code the CRUD (Create, Read, Update, Delete) operations (see Figure 2) for the Person table using Node.js, express, a REST API, and the mysql2 library. You can use the code from lecture 06 (pages 42 to 54) as a starting point. Make sure to test your code by running it and checking the results in the database.



ID	First	Last	Email	Birthdate	Actions
1	John	Doe	john.doe@example.com	1989-12-31	Delete Update
3	Alice	Johnson	alice.johnson@example.com	1992-09-29	Delete Update
4	Bob	Brown	bob.brown@example.com	1988-12-19	Delete Update
7	David	Miller	david.miller@example.com	1987-11-04	Delete Update
8	Sarah	Taylor	sarah.taylor@example.com	1993-02-17	Delete Update
10	Olivia	Thomas	olivia.thomas@example.com	1994-04-21	Delete Update
11	James	Jackson	james.jackson@example.com	1986-10-29	Delete Update
12	Sophia	White	sophia.white@example.com	1990-06-14	Delete Update
13	William	Harris	william.harris@example.com	1988-01-04	Delete Update
14	Isabella	Martin	isabella.martin@example.com	1992-11-19	Delete Update
15	Benjamin	Garcia	benjamin.garcia@example.com	1987-04-09	Delete Update
16	Mia	Clark	mia.clark@example.com	1993-09-04	Delete Update
17	Jacob	Rodriguez	jacob.rodriguez@example.com	1989-12-24	Delete Update
18	Charlotte	Lewis	charlotte.lewis@example.com	1991-03-14	Delete Update
19	Ethan	Lee	ethan.lee@example.com	1986-07-29	Delete Update
20	Amelia	Walker	amelia.walker@example.com	1994-05-09	Delete Update
22	Evelyn	Allen	evelyn.allen@example.com	1990-02-19	Delete Update
28	Jeremy	TheLandre	jtl@google.com	2001-07-26	Delete Update

Figure 2: The CRUD web page…

4 If you are done…

In case you are done with the workshop very fast, you can add two other tables and add a REST API to play with the JOIN of the tables. :) B+